Sir Syed University of Engineering & Technology
Faculty of Computing and Applied Sciences
Department of Computer Science & Information Technology

Mid Term Examinations (Fall 2022)

Course Code with Title	**CS-215T: Computer Organization & Assembly Language**		**Program**	Bachelor of Computer Science BS(CS)
Instructor(s)	**Mr. Shakir Karim / Dr. Abdul Hameed Pitafi**		**Semester**	3^{rd}
Start date & Time	December 09, 2022 at 11:15 AM	**End Time**	12:45 PM	
Maximum Marks	30			

08549

IMPORTANT INSTRUCTIONS:

Read the following Instructions carefully:

- Attempt All Questions.
- Use of calculator (simple/scientific/programmable) is not allowed.
- Mobile Phones are strictly prohibited in examination premises.

Q.1. **(03)**

Illustrate the computer model proposed by von Neumann using block diagram.

Q.2. **(05)**

Show step-by-step standard process of signed multiplication to multiply (-5) and (+6). Assume 4-bits registers to hold signed numbers.

0100 0010 1111 0010 1000 0000 000

-30

42F23000H

Q.3. **(05)**

Represent **121.0625_{10}** in **IEEE 754 single precision** format. Show all the necessary steps involved in the process.

Q.4. **(03+03+03)**

Consider that **DS = 70ACh, SS = 850Fh, BP = 1000h, SP = FFFDh, AX = 1734h, BX = 5678h,** and **DI = 560Dh.** Identify the addressing modes for each of the following instructions and determine the memory addresses, contents of memory location after execution.

(i) **MOV AL, [BP+DI]** (ii) **PUSH DI** (iii) **MOV [1254H], BX**

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Q.5. **(02+02+02+02)**

Refer to the table 1 and instruction set summary (for *MOV* and *ADD* instructions), write machine code for the given instructions.

(i) **ADD [BP+SI], AL**

(ii) **MOV [DI]+1234H, DL**

(iii) **MOV DS, 5434H**

(iv) **ADD [DI], BL**

Table 1: R/M and MOD values

R/M \ MOD	00 0-bit displacement	01 8-bit displacement	10 16-bit displacement	11 w=0	11 w=1
000	[BX+SI]	[BX+SI]+D8	[BX+SI]+D16	AL	AX
001	[BX+DI]	[BX+DI]+D8	[BX+DI]+D16	CL	CX
010	[BP+SI]	[BP+SI]+D8	[BP+SI]+D16	DL	DX
011	[BP+DI]	[BP+DI]+D8	[BP+DI]+D16	BL	BX
100	[SI]	[SI]+D8	[SI]+D16	AH	SP
101	[DI]	[DI]+D8	[DI]+D16	CH	BP
110	DIRECT	[BP]+D8	[BP]+D16	DH	SI
111	[BX]	[BX]+D8	[BX]+D16	BH	DI

8088

8086/8088 Instruction Set Summary

Mnemonic and Description	Instruction Code			
DATA TRANSFER				
MOV = Move:	7 6 5 4 3 2 1 0	7 6 5 4 3 2 1 0	7 6 5 4 3 2 1 0	7 6 5 4 3 2 1 0
Register/Memory to/from Register	1 0 0 0 1 0 d w	mod reg r/m		
Immediate to Register/Memory	1 1 0 0 0 1 1 w	mod 0 0 0 r/m	data	data if w = 1
Immediate to Register	1 0 1 1 w reg	data	data if w = 1	
Memory to Accumulator	1 0 1 0 0 0 0 w	addr-low	addr-high	
Accumulator to Memory	1 0 1 0 0 0 1 w	addr-low	addr-high	
Register/Memory to Segment Register	1 0 0 0 1 1 1 0	mod 0 reg r/m		
Segment Register to Register/Memory	1 0 0 0 1 1 0 0	mod 0 reg r/m		
ARITHMETIC	7 6 5 4 3 2 1 0	7 6 5 4 3 2 1 0	7 6 5 4 3 2 1 0	7 6 5 4 3 2 1 0
ADD = Add:				
Reg./Memory with Register to Either	0 0 0 0 0 0 d w	mod reg r/m		
Immediate to Register/Memory	1 0 0 0 0 0 s w	mod 0 0 0 r/m	data	data if s:w = 01
Immediate to Accumulator	0 0 0 0 0 1 0 w	data	data if w = 1	